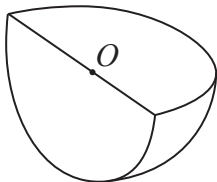


19 Shape S is one quarter of a solid sphere, centre O.



Shape S

Volume of sphere = $\frac{4}{3}\pi r^3$

Surface area of sphere = $4\pi r^2$

The volume of S is $576\pi\text{ cm}^3$

Find the surface area of S.

Give your answer correct to 3 significant figures.

You must show your working.

.....cm²